

```
In [ ]: #Print Patterns in python
```

```
In [ ]: #1. Pattern 1
```

```
In [1]: for i in range(1,5):  
        i = i+1  
        print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [2]: for i in range(1,5):  
        print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [3]: for i in range(1,5):  
        if i<=5:  
            print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [4]: for j in range(4):  
        print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [6]: for i in range(4):  
        for j in range(4):
```

```
    print("#", end = ' ')
    print()
```

```
# # # #
# # # #
# # # #
# # # #
```

In []: *#1. Pattern 2*

```
In [7]: for i in range(4):
        for j in range(i+1):
            print("#", end = ' ')
        print()
```

```
#
# #
# # #
# # # #
```

```
In [8]: for i in range(1,5):
        print("# " * i)
```

```
#
# #
# # #
# # # #
```

```
In [9]: for i in range(1,5):
        for j in range(4):
            if i>j:
                print("#", end = ' ')
        print()
```

```
#
# #
# # #
# # # #
```

```
In [13]: for i in range(1,5):
        for j in range(i):
            print("#", end = ' ')
        print()
```

```
#
# #
# # #
# # # #
```

In []: *#3. Pattern 3*

```
In [14]: for i in range(4):
          for j in range(4-i):
              print("#", end = ' ')
          print()
```

```
# # # #
# # #
# #
#
```

```
In [16]: for i in range(1,5):
          print("# " * (5-i))
```

```
# # # #
# # #
# #
#
```

In []: *#More Patterns:*

```
In [17]: #1. Right Angle Triangle Pattern:
          for i in range(1,6):
              print(' ' * ' * i)
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [18]: #2. Inverted Right Angle Triangle Pattern:
          for i in range(5,0,-1):
              print(' ' * ' * i)
```

```

* * * * *
* * * *
* * *
* *
*

```

```

In [25]: #3. Pyramid Pattern:
for i in range(1,6):
    print('*(5-i)+' * '*(2*i-1))

```

```

*
* * *
* * * * *
* * * * * * *
* * * * * * * *

```

```

In [26]: for i in range(1,6):
          print(' * '*(2*i-1))

```

```

*
* * *
* * * * *
* * * * * * *
* * * * * * * *

```

```

In [27]: #4. Inverted Pyramid Pattern:
for i in range(5,0,-1):
    print('*(5-i)+' * '*(2*i-1))

```

```

* * * * * * * *
* * * * * *
* * * *
* * *
*

```

```

In [28]: for i in range(5,0,-1):
          print(' * '*(2*i-1))

```

```

* * * * * * * *
* * * * * *
* * * *
* * *
*

```

```
In [29]: #5. Diamond Pattern:
for i in range(1,6):
    print('*'(5-i)+' * '*'(2*i-1))
for i in range(4,0,-1):
    print('*'(5-i)+' * '*'(2*i-1))
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * * *
* * * * * * *
* * * * *
* * *
*
```

```
In [31]: #6. Hallow Square Pattern:
for i in range(5):
    for j in range(5):
        if i==0 or i==4 or j==0 or j==4:
            print('*', end = ' ')
        else:
            print(' ', end = ' ')
    print()
```

```
* * * * *
*       *
*       *
*       *
* * * * *
```

```
In [35]: #7. Full Square Pattern:
for i in range(5):
    print('*' * 5)
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [46]: #8. Right Angle triangle (Number Pattern):
for i in range(1,6):
    print(' '.join(str(x) for x in range(1,i+1)))
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
In [45]: #9. Inverted Right Angle triangle (Number Pattern):
for i in range(5,0,-1):
    print(' '.join(str(x) for x in range(1,i+1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
In [44]: #10. Floyd's triangle:
num = 1
for i in range(1,6):
    for j in range(1,i+1):
        print(num, end = ' ')
        num+=1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
In [43]: #11. Hollow Right Angle Triangle:
for i in range(1,6):
    for j in range(1,i+1):
        if j == 1 or j == i or i == 5:
            print('*', end = ' ')
        else:
            print(' ', end = ' ')
    print()
```

```

*
*  *
*   *
*    *
*   *  *

```

In [58]: #12. Hallow Pyramid Pattern:

```

for i in range(1,6):
    for j in range(5-i):
        print(' ',end = ' ')
        for j in range(2*j-1):
            if j == 0 or j == 2*i-2 or i == 5:
                print(' * ', end = ' ')
            else:
                (' ', end = ' ')
        print()

```

Cell In[58], line 9
 (' ', end = ' ')

SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?

In [73]: #13. Hallow Diamond Pattern:

```

n = 5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(2*i-1):
        if j == 0 or j == 2*i-2:
            print(' * ', end = ' ')
        else:
            print(' ', end = ' ')
    print()

for i in range(n-1,0,-1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(2*i-1):
        if j == 0 or j == 2*i-2:
            print(' * ', end = ' ')
        else:

```

```

        print(' ', end = ' ')
print()

```

```

      *
    *  *
  *      *
*          *
*        *
  *      *
    *  *
      *

```

In [75]: #14. Hallow Diamond (Number Pattern):

```

n = 5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(2*i-1):
        if j == 0 or j == 2*i-2:
            print( i , end = ' ')
        else:
            print(' ', end = ' ')
    print()

for i in range(n-1,0,-1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(2*i-1):
        if j == 0 or j == 2*i-2:
            print( i , end = ' ')
        else:
            print(' ', end = ' ')
    print()

```



```

      1
    2 2
  3   3
4    4
5      5
  4    4
    3   3
      2 2
        1

```

In [105...

```

#15. Butterfly Pattern:
n = 5
for i in range(1,n+1):
    for j in range(1, i+1):
        print(j, end = ' ')
    for j in range(2*(n-i)):
        print(' ', end = ' ')
    for j in range(1, i+1):
        print(j, end = ' ')
    print()

for i in range(n,0,-1):
    for j in range(1, i+1):
        print(j, end = ' ')
    for j in range(2*(n-i)):
        print(' ', end = ' ')
    for j in range(1, i+1):
        print(j, end = ' ')
    print()

n = 5
for i in range(1,n+1):
    for j in range(i):
        print('*', end = ' ')
    for j in range(2*(n-i)):
        print(' ', end = ' ')
    for j in range(i):
        print('*', end = ' ')
    print()

for i in range(n,0,-1):
    for j in range(i):

```

```

    print('*', end = ' ')
    for j in range(2*(n-i)):
        print(' ', end = ' ')
    for j in range(i):
        print('*', end = ' ')
    print()

```

```

1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1
*                *
* *              * *
* * *            * * *
* * * *          * * * *
* * * * * * * * * *
* * * * * * * * * *
* * * *          * * * *
* * *            * * *
* *              * *
*                *

```

In [90]: #16. Hallow Number Pyramid:

```

n = 5
for i in range(1, n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(1, 2*i):
        if j == 1 or j == 2*i-1 or i == n:
            print(i, end = ' ')
        else:
            print(' ', end = ' ')
    print()

```

```

      1
    2  2
  3      3
4      4
5 5 5 5 5 5 5 5

```

In [91]: #17. Full Star Pyramid :

```

n = 5
for i in range(1, n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(1, 2*i):
        print(' * ', end = ' ')
    print()

```

```

      *
    * * *
  * * * * *
* * * * * * *

```

In [92]: #18. Inverted Full Star Pyramid :

```

n = 5
for i in range(n, 0, -1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(2*i-1):
        print(' * ', end = ' ')
    print()

```

```

* * * * *
 * * * *
  * * *
   * *
    *

```

In [96]: #19. Left Aligned Pyramid Pattern :

```

n = 5
for i in range(1, n+1):
    for j in range(i):
        print('* ', end = ' ')
    print()

```

```

for i in range(1, n+1):
    for j in range(1,i+1):
        print( j , end = ' ')
    print()

```

```

*
* *
* * *
* * * *
* * * * *
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

In [107... *#20. Right Aligned Pyramid Pattern :*

```

n = 5
for i in range(1, n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(1,i+1):
        print( j , end = ' ')
    print()

for i in range(1, n+1):
    for j in range(n-i):
        print(' ', end = ' ')
    for j in range(i):
        print('*', end = ' ')
    print()

```

```

      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
   *
  * *
 * * *
* * * *
* * * * *

```

